

Network Basics & Security Documentation

Purpose

This document provides a basic overview of networking concepts, network components, protocols, and basic security controls. It is intended to document technical information in a clear and structured manner.

Scope

This document covers basic networking concepts, OSI and TCP/IP models, common network devices, protocols and ports, basic network security controls, and identified security observations.

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Network Diagram

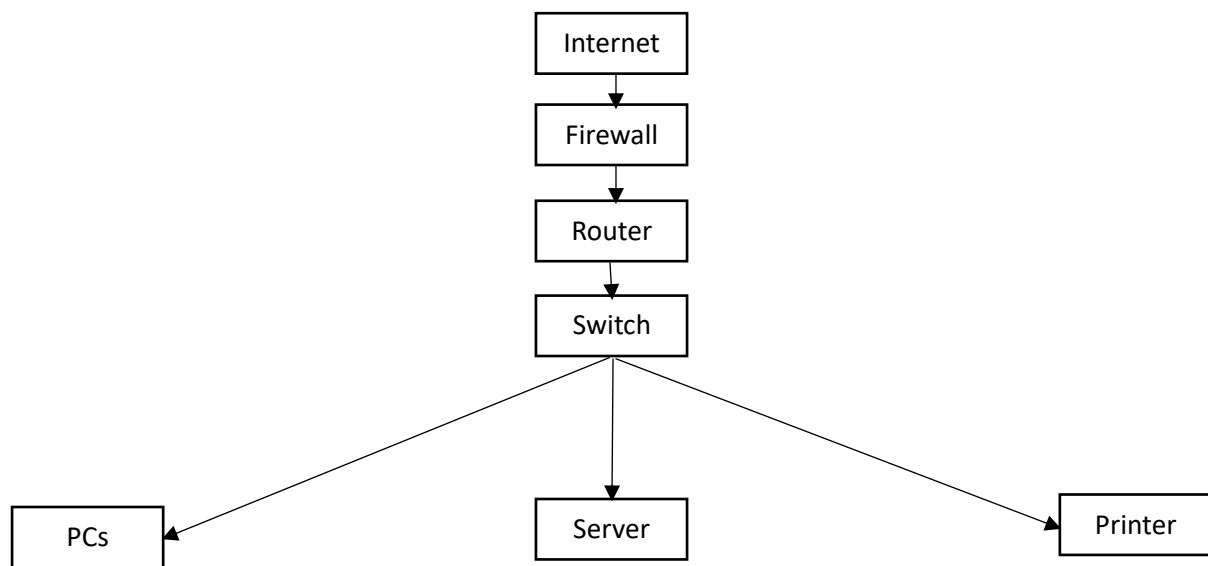


Figure 1: Basic Network Diagram

OSI/TCP-IP Overview

OSI Model

The OSI (Open Systems Interconnection) model is a seven-layer conceptual model used to understand how data communication occurs between network devices.

Layer	Name	Basic function
7	Application	Provides network services to applications
6	Presentation	Handles data format, encryption, and compression
5	Session	Establishes and manages communication sessions
4	Transport	Provides end-to-end data delivery
3	Network	Handles logical addressing and routing
2	Data Link	Handles frames and MAC addresses
1	Physical	Transmits data as signals over the physical medium

TCP/IP Model

The TCP/IP model is a practical networking model used to describe how devices communicate over networks and the Internet.

Layer	Basic Functions
Application	Provides network services and protocols to applications
Transport	Provides communication between applications and handles data delivery
Internet	Handles logical addressing and routing
Network Access	Handles communication over the physical network

OSI vs TCP/IP

OSI Model	TCP/IP Model
7 layers	4 layers
Conceptual/reference model	Practical networking model
Separates Session and Presentation layers	Includes them within Application
Separates Data Link and Physical layers	Combines them into Network Access

Important Networking Concepts

IP Address

An IP address is a logical address used to identify a device on a network and enable communication between devices.

IPv4

IPv4 is a version of the Internet Protocol that uses 32-bit addresses to identify devices on a network.

IPv6

IPv6 is a newer version of the Internet Protocol that uses 128-bit addresses to provide a much larger address space.

Public vs Private IP

A Public IP is used to communicate over the Internet, while a Private IP is used within an internal network.

MAC Address

A MAC address is a unique hardware-level address associated with a network interface and is used for communication within a local network.

Subnet Mask

A subnet mask determines which portion of an IP address represents the network and which portion identifies the device.

Default Gateway

A default gateway is the device, usually a router, that forwards traffic from a local network to other networks.

DNS

DNS (Domain Name System) translates human-readable domain names into IP addresses.

DHCP

DHCP (Dynamic Host Configuration Protocol) automatically assigns network configuration such as IP addresses to devices.

HTTP

HTTP (Hypertext Transfer Protocol) is used to transfer information between web browsers and web servers.

HTTPS

HTTPS is the secure version of HTTP that uses encryption to protect data exchanged between a client and server.

SSL

SSL (Secure Sockets Layer) is an older security protocol designed to protect data transmitted over a network. It has been replaced by TLS.

TLS

TLS (Transport Layer Security) is a security protocol that provides encryption and secure communication over networks.

NAT

NAT (Network Address Translation) allows multiple devices on a private network to share a public IP address when accessing external networks.

Ports

Ports are logical communication endpoints used to identify specific services or applications running on a device.

SSH

SSH is a protocol used for secure remote access and administration of networked systems.

FTP

FTP (File Transfer Protocol) is used to transfer files between a client and a server over a network.

SFTP

SFTP (SSH File Transfer Protocol) provides secure file transfer over SSH.

VPN

A VPN (Virtual Private Network) creates a secure connection over a network and helps protect data in transit.

Proxy Server

A proxy server acts as an intermediary between a user and another network or service, forwarding requests on the user's behalf.

Common Protocols and Ports

Protocol	Port	Purpose
HTTP	80	Used for web communication
HTTPS	443	Secure web communication
DNS	53	Resolves domain names to IP addresses
DHCP	67/68	Automatically assigns IP configuration
SSH	22	Secure remote access
FTP	20/21	File transfer
SFTP	22	Secure file transfer
SMTP	25	Sending email
POP3	110	Receiving email
IMAP	143	Accessing email from a mail server

Common Network Devices

Router

A router connects different networks and forwards data packets between them.

Switch

A switch connects devices within a local network and forwards data to the appropriate device.

Hub

A hub connects multiple devices in a network and broadcasts received data to all connected devices.

Firewall

A firewall monitors and controls incoming and outgoing network traffic based on defined security rules.

Access Point

An access point allows wireless devices to connect to a wired network using Wi-Fi.

Server

A server provides services, resources, or data to other devices connected to a network.

Endpoint

An endpoint is a device such as a laptop, desktop, or mobile device that connects to a network and is used by an end user.

Network Device Comparison

Device	Main Function	Common Use
Router	Connects different networks and forwards traffic	Connects a local network to the Internet
Switch	Connects devices within a local network	Connects PCs, servers, and printers
Hub	Broadcasts data to all connected devices	Basic/older network setups
Firewall	Controls network traffic based on security rules	Protects networks from unauthorized traffic
Access Point	Provides wireless network connectivity	Connects Wi-Fi devices to a network
Server	Provides services or resources to other devices	File, web, or application services
Endpoint	Device used by an end user	Laptop, desktop, or mobile device

Basic Network Security Controls

Firewall

A firewall monitors and controls incoming and outgoing network traffic based on defined security rules. It helps prevent unauthorized network access.

IDS

An IDS (Intrusion Detection System) monitors network or system activity and detects suspicious or potentially malicious activity.

IPS

An IPS (Intrusion Prevention System) detects suspicious activity and can automatically take action to block or prevent threats.

VPN

A VPN (Virtual Private Network) creates an encrypted connection over a network to help protect data during transmission.

Network Segmentation

Network segmentation divides a network into separate sections to limit access and reduce the impact of a security incident.

Access Control

Access control ensures that only authorized users or devices can access specific network resources.

Security Control Summary

Security Control	Purpose	Security Benefit
Firewall	Controls network traffic	Blocks unauthorized traffic
IDS	Detects suspicious activity	Helps identify potential attacks
IPS	Detects and blocks threats	Helps prevent malicious activity
VPN	Encrypts network communication	Protects data in transit
Network Segmentation	Separates network areas	Limits security impact
Access Control	Restricts resource access	Prevents unauthorized access

Security Observations / Risks

The following observations identify common security risks that may exist in a basic network environment and the potential impact they could have on security.

Observation	Risk	Potential Impact	Recommended Control
Unrestricted network access	Unauthorized users may access network resources	Data exposure or unauthorized activity	Implement access control and firewall rules
Outdated network devices	Known vulnerabilities may remain unpatched	Attackers may exploit vulnerabilities	Regularly update and patch network devices
Lack of network segmentation	An attacker may move between network areas	Increased impact of a security incident	Implement network segmentation
Weak remote access security	Unauthorized users may gain remote access	Account compromise or data exposure	Use VPN and strong authentication
No network monitoring	Suspicious activity may go undetected	Delayed detection of security incidents	Implement IDS/IPS and monitoring

Version History

Version	Date	Author	Changes
1.0	10 August 2026	Durva Amin	Initial Release

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